

16 A

Acceleration of SHM given by

$$a = -a_0 \sin(\omega t)$$

Displacement of SHM given by

$$x = x_0 \sin(\omega t)$$

Ratio

$$\begin{aligned}\frac{x_{t=\frac{5T}{8}}}{x_0} &= \frac{x_0 \sin\left(\frac{2\pi}{T} \times \frac{5T}{8}\right)}{x_0} \\ &= x_0 \sin\left(\frac{5\pi}{4}\right) \\ &= -0.71\end{aligned}$$